

To achieve a fully automated exception management system for Clox, you must transition from manual dispatcher intervention to a **"Control Loop" architecture**. In this model, the system detects, evaluates, and resolves driver unavailability without human input.

1. The Automated Exception Loop

For an end-to-end automated flow, implement the following steps within the Clox platform:

- **Step A: In-App "Exception" Trigger:** When a driver reports they are unavailable, they must interact with a structured "Status Update" or "Breakdown" menu in the driver app. This replaces informal text messages with system-readable data (e.g., reason codes like *Vehicle Issue, Personal Emergency, Not Starting Shift*).
- **Step B: Automatic "Force-Offline" & Audit:** Upon selecting "Unavailable," the system must immediately toggle the driver's status to "Inactive." This prevents the platform from assigning further jobs until the driver performs a "Re-verification" or "Ready for Duty" check.
- **Step C: Intelligent Re-Optimization:** The platform should trigger a re-optimization algorithm that automatically searches for the next best-matched driver based on:
 - **Proximity:** GPS-based location to the pickup point.
 - **Capacity/Skills:** Matching vehicle type and service requirements.
 - **Availability:** Current shift status and remaining hours.
- **Step D: Automated Stakeholder Updates:**
 - **Customer:** Automatically notify the sender: *"Your driver has updated their status; we have automatically assigned a new driver to ensure your delivery stays on track"*.
 - **Carrier Admin:** Send a summary notification to the fleet company's portal regarding the cancellation for their own internal performance review.

2. Required Platform Features

To support this automation, your platform needs these core technical capabilities:

Feature	Functionality
Reason Codes	Categorizes why a driver is unavailable for analytics.
Constraint-Based Logic	Ensures re-assignment respects vehicle capacity and timing constraints.
Webhook/API Triggers	Automatically pushes updates to customer portals and mobile notifications.

Performance Scoring

Automatically tracks cancellation rates against "Reliability Thresholds".

3. Implementing the "Carrier Liability" Logic

Since Clox works with third-party carriers, the automation should reflect your business agreements:

1. **Immediate Re-assignment (Intra-fleet):** Attempt to re-assign the job within the *same* carrier's fleet first.
2. **Marketplace Escalation (Inter-fleet):** If no driver from the original carrier accepts the rescue within a defined threshold (e.g., 5 minutes), the job automatically opens to the wider Clox marketplace.
3. **Financial Adjustment:** The platform's smart contracts should automatically re-allocate the service fee to the new driver/carrier while maintaining the original customer price.

Recommendation for Clox

To make this truly "hands-off," avoid creating a "decision-pending" state for your staff. Set up your system to **always** attempt an automated resolution first. Only if the re-assignment fails (e.g., no drivers available in the entire radius) should the system generate a high-priority "Dispatch Intervention" alert for your team.

This ensures your team only manages truly complex scenarios, while routine unavailability is handled in real-time, preserving the customer's experience.